Shaping the 6G future
From vision to action

Powered by
Nokia Technology Training & Certifications

Shaping the 6G Future

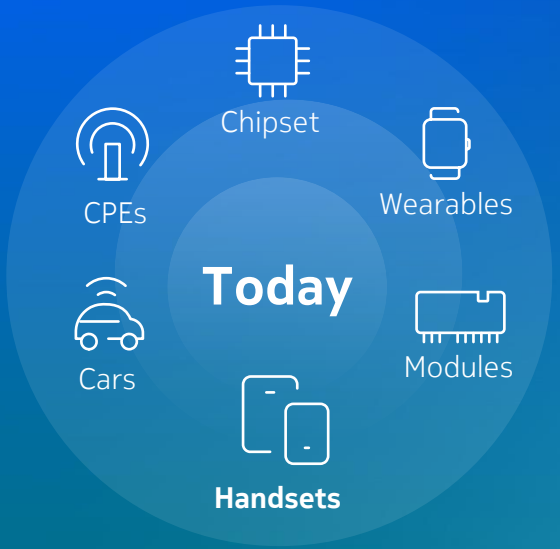
1	**Key trends shaping the ecosystem**
2	Why 6G, and its design principles
3	6G vision, use cases and technology enablers
4	Building a tangible foundation for 6G

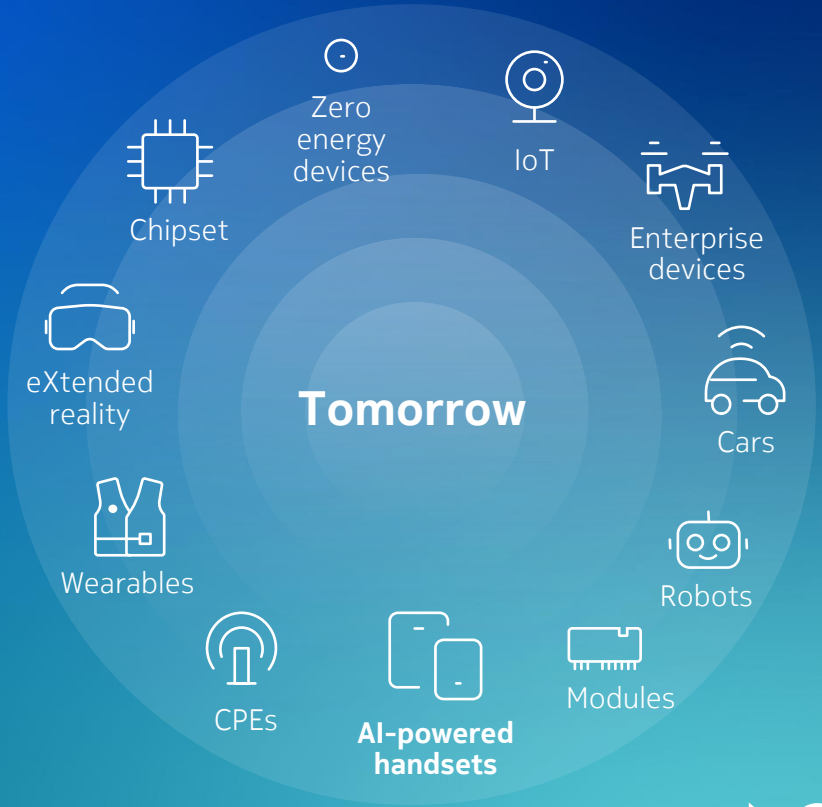
Trend #1 – As 5G adoption continues to scale, it will demand higher network performance

Existing	New	Emerging / 5G-Advanced
Subscription conversion 5G conversion	**Fixed wireless access** Home & enterprise	**Critical Communication** Utility, defence and enterprise
Unlimited 5G data package All you can eat	**5G enterprise service** Network slicing	**Event-based XR experience** Event services
Speed-based data package Need for speed	**5G AR/VR experience** Service bundle with slicing	**Network as Service** Programmable Networks
		Latency critical computing Edge compute

© 2024 Nokia

Trend #2 - Device ecosystem is expanding





© 2024 Nokia

Trend #3 - Adoption of emerging technologies impacts our networks

AI is expected to contribute around $450 billion in value across multiple industries by 2030.

ABI Research

92% of business investments in APIs will rise or remain the same during the next year.

Postman API Report

86% of organizations report an increase in cloud initiatives.

Accenture

The estimated value of the worldwide digital twin market in 2026 is $48 billion (58% CAGR from 2020 to 2026).

Siemens Research

74% companies using autonomous & remote-controlled operations see sustainability as a key benefit

Nokia and EY study

© 2024 Nokia

Shaping the 6G Future

1	Key trends shaping the ecosystem
2	**Why 6G, and its design principles**
3	6G vision, use cases and technology enablers
4	Building a tangible foundation for 6G

The push and pull trends point to the need for 6G

Trends and challenges

Demand for higher network performance

The vastly growing device ecosystem

The surging power of AI and emergence of the API economy

How 6G can address them

Design principles for 6G

Sustainable

Resilient & Secure

Digital inclusive

© 2024 Nokia

Design principles for 6G

Sustainable

- 10X capacity increase with 50% power reduction, compared to 5G

Resilient & Secure

- Increasing security and privacy risks require higher levels of control

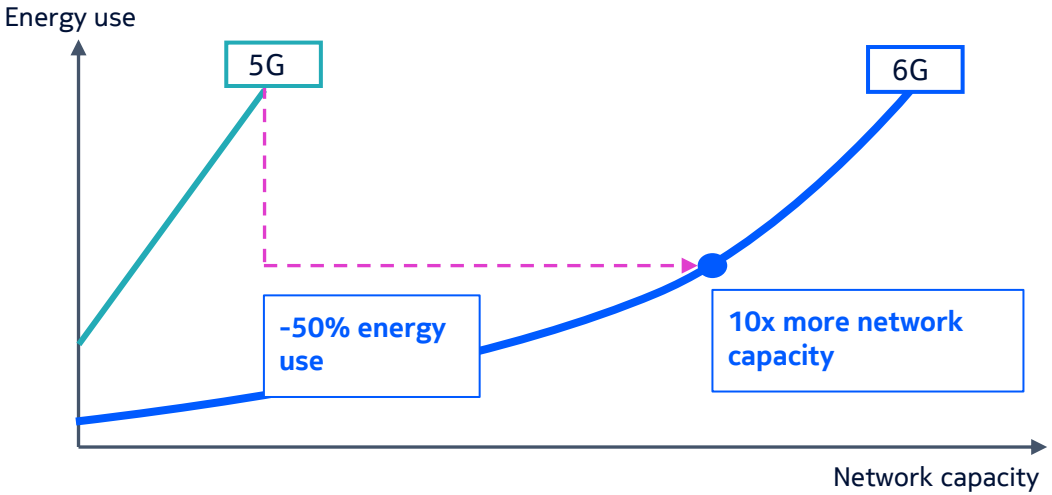
Digital inclusive

- Aims to address three key factors: accessibility, affordability and consumability

© 2024 Nokia

Green by design

Lower the baseline power requirements of the network at all times



Minimize relative power consumption in low-to-mid-load conditions: **zero users – zero power consumption**

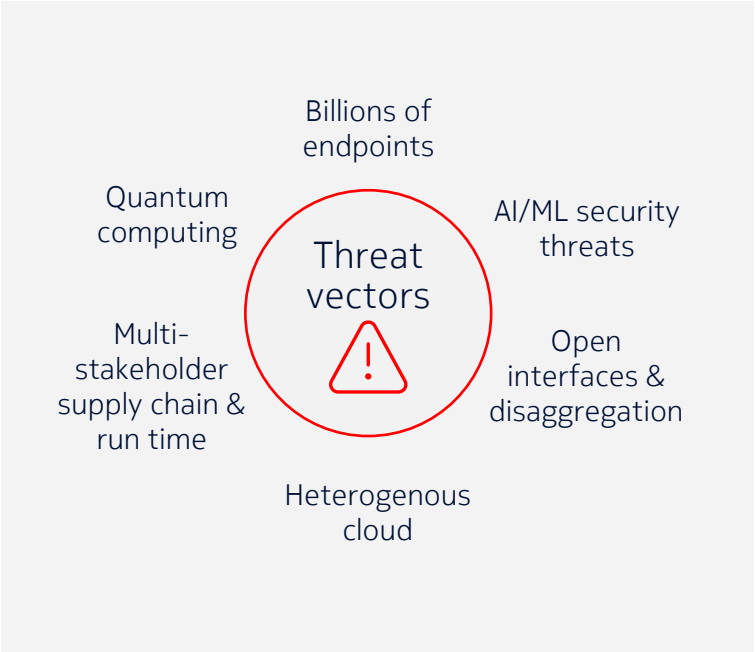
Maximize the efficiency of the network in mid-to-high-volume conditions: **substantial reduction in energy per bit**

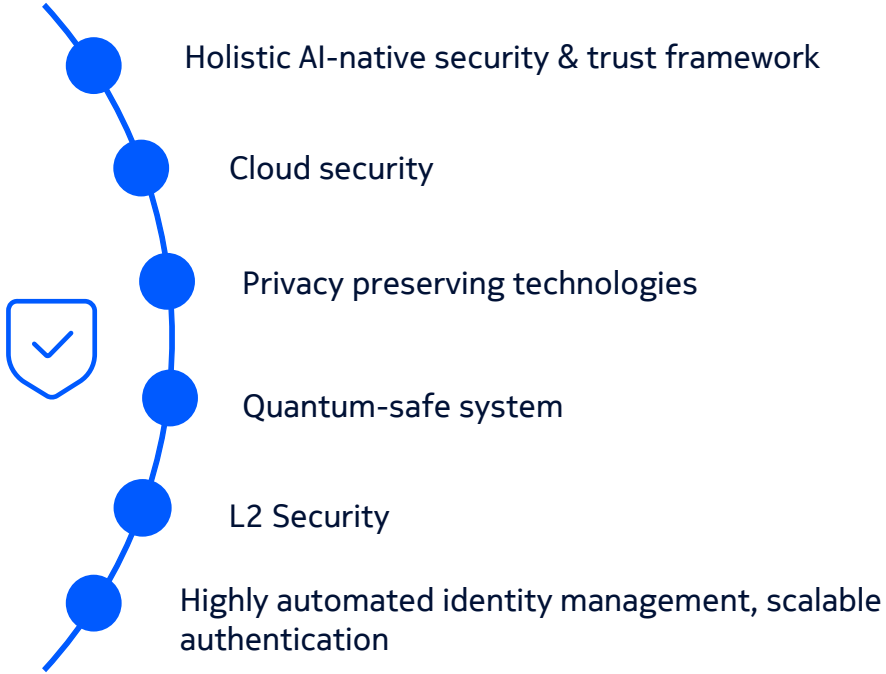
6G target: 10x more capacity with 50% less power usage
➔ 20x lower kW/Gbps

© 2024 Nokia

Security and privacy

Design new solutions to tackle emerging threat vectors





 © 2024 Nokia

Digital inclusion

Accessibility

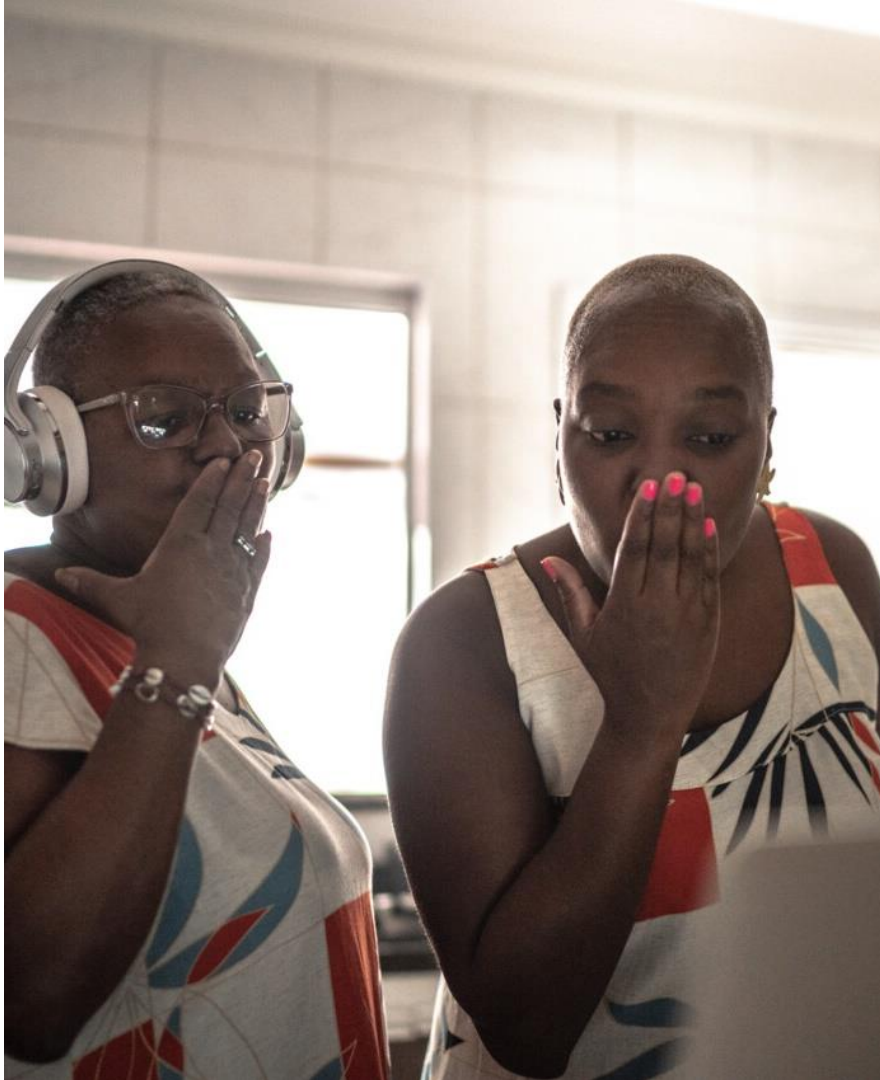
- New solutions such as integrated non-terrestrial networks and terrestrial networks will help make global connectivity a reality

Affordability

- Interoperability from global standards and delivery of cost-effective 6G solutions

Consumability

- User-centric design with enhanced usability, serviceability and operability, ensuring intuitive human-machine interactions



 © 2024 Nokia

Shaping the 6G Future

1	Key trends shaping the ecosystem
2	Why 6G, and its design principles
3	**6G vision, use cases and technology enablers**
4	Building a tangible foundation for 6G

The human, physical and digital worlds will become seamlessly integrated in the 6G era

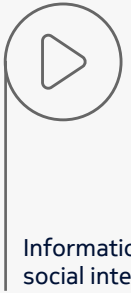
1991 & 2001

2G & 3G

Connections and mobility

2009

4G



Information and social interaction

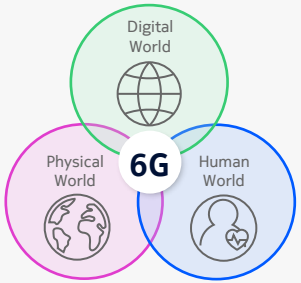
2019

5G

Machine and sensor data

2030 & beyond

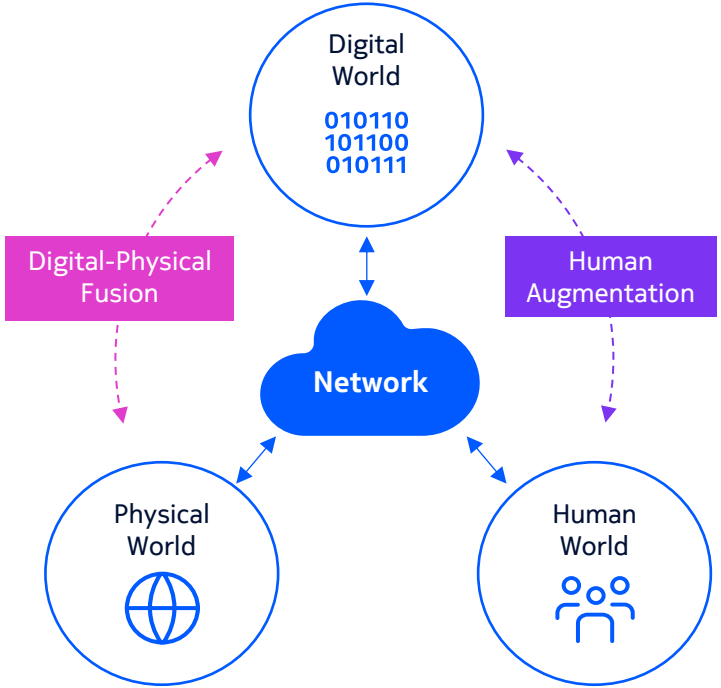
6G



Connecting the human, physical and digital worlds to augment the potential of human beings

© 2024 Nokia

The 6G era will be defined by digital-physical fusion and human augmentation



Digital-Physical Fusion

Dynamic, network-connected representations of real-world things in the digital world

Allows the physical world to be replicated, simulated and automated within the digital world - opening doors to a wealth of new possibilities benefiting humanity.

Human Augmentation

Extensions that enable people to interact with and within the digital world

Evolves the digital world from a source of two-dimensional experiences and information, to a focal point of immersive and productive interaction

 © 2024 Nokia

Realization of this 6G vision comes in stages

Technology enablers and services examples

6G day-one focus

- Extreme MIMO on existing grid
- Smooth migration and core evolution
- Programmable networks and API native
- Framework for native AI
- Framework for energy efficiency
- Non-Terrestrial Networks (NTN) support for all UE types

NextG mobile broadband

Fixed Wireless Access (FWA)

Immersive /Cloud gaming

Extended Reality

IoT/LPWA native support

"Integrated" global connectivity

6G to build on 5G success and do so in a more efficient, economical, scalable and sustainable way

6G evolution and beyond

The human, physical and digital worlds will become seamlessly integrated in the 6G era

- From connectedness to **togetherness:** immersive holographic experience, connect the unconnected

- From information to **knowledge:** cognitive and complete context awareness, leveraging ambient IoT, digital twins, sensing

- From efficiency to **purpose:** mission & life-critical services supported by subnetworks

Augmenting humanity and the full realization of digital-physical fusion

© 2024 Nokia

6G Day One focus

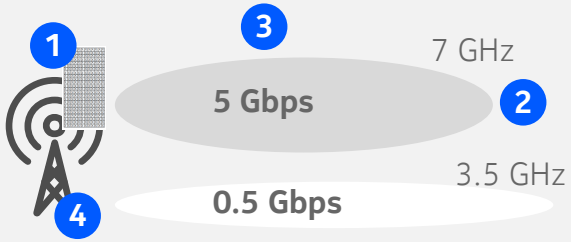
The sustainable answer to the unceasing demand for connectivity

- Extreme MIMO on existing grid
- Smooth migration and core evolution
- Programmable networks and API native
- Framework for native AI
- Framework for energy efficiency
- Non-Terrestrial Networks

© 2024 Nokia

Extreme massive MIMO supported from existing RF site grid

Targets for 6G deployments at 6.4 – 8.4 GHz



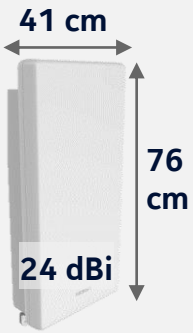
1 = Reuse of existing base station sites for new bands

2 = Similar coverage with 7 GHz as with 3.5 GHz

3 = Cell capacity 10x compared to 3.5 GHz

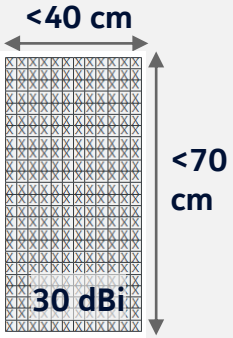
4 = Energy efficiency 2x at low load

5G at 3.5 GHz



8x12x2=192

6G at 6-8 GHz



16x24x2=768

Coverage solution: 4x **more antenna elements** in the same physical area to boost antenna gain up to 6 dB

Capacity solution: 4x **more bandwidth** and 3x more spectral efficiency with **more TRXs & beamforming**

Energy efficiency solution: enhanced sleep mode and **very fast deep sleep**

© 2024 Nokia

A single architecture for simple evolution

5G today

7 architectures defined in 3GPP and 2 implemented (NSA, SA)

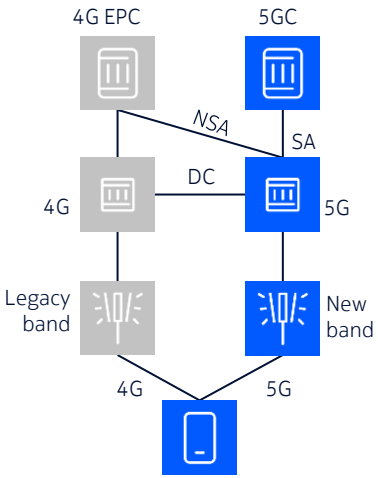
5G core is a major upgrade on top of 4G core

4G + 5G dual connectivity (DC) and 4G RAN – 5G RAN interface

Two radios (LTE + 5G) active in devices simultaneously

Refarming mostly with static spectrum split

4G to 5G migration



6G evolution

Only **single architecture** option for simple evolution

6G **core is a smooth migration** from 5G core

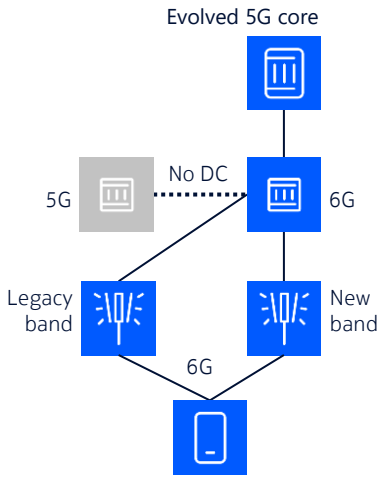
No 5G – 6G dual connectivity (just MRSS* control)

Only a single radio active (6G) in a device at a time

Multi-Radio Spectrum Sharing (MRSS) for truly dynamic refarming

** MRSS: Multi-RAT spectrum sharing*

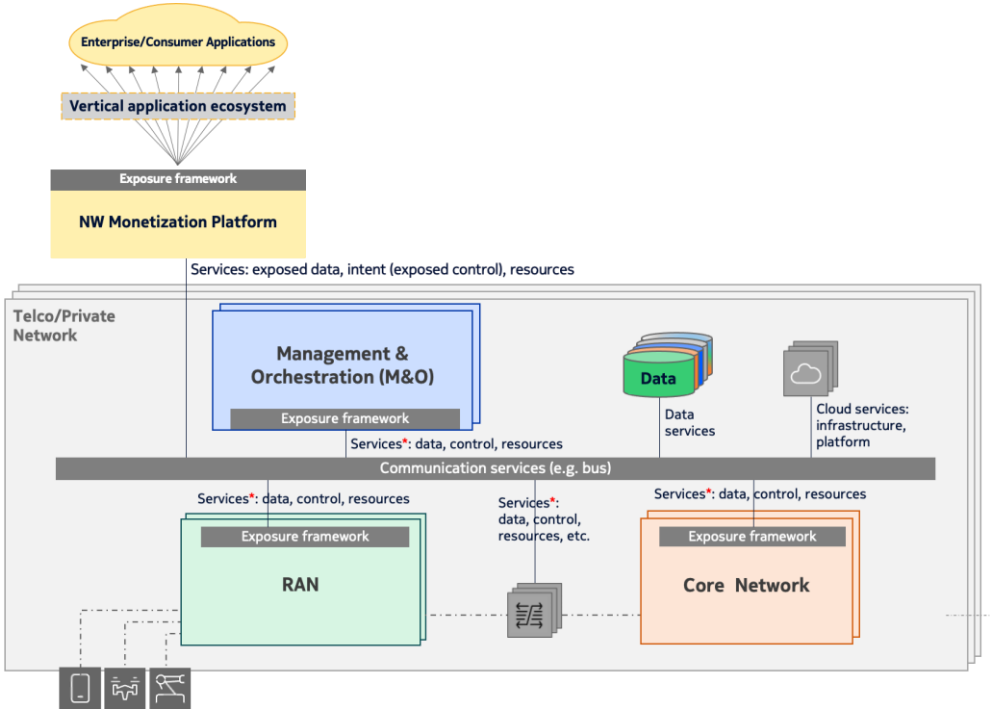
5G to 6G migration



© 2024 Nokia

Programmable network and API native

Fostering innovation and promoting monetization



- Streamlined, unified exposure architecture for consistent consumer experience
- Holistic support of use cases, following a top-down approach
- Enhanced, governed access to invaluable data to fuel innovation
- Flexible intelligence distribution, with inherent exposure to enriching information within and outside the network

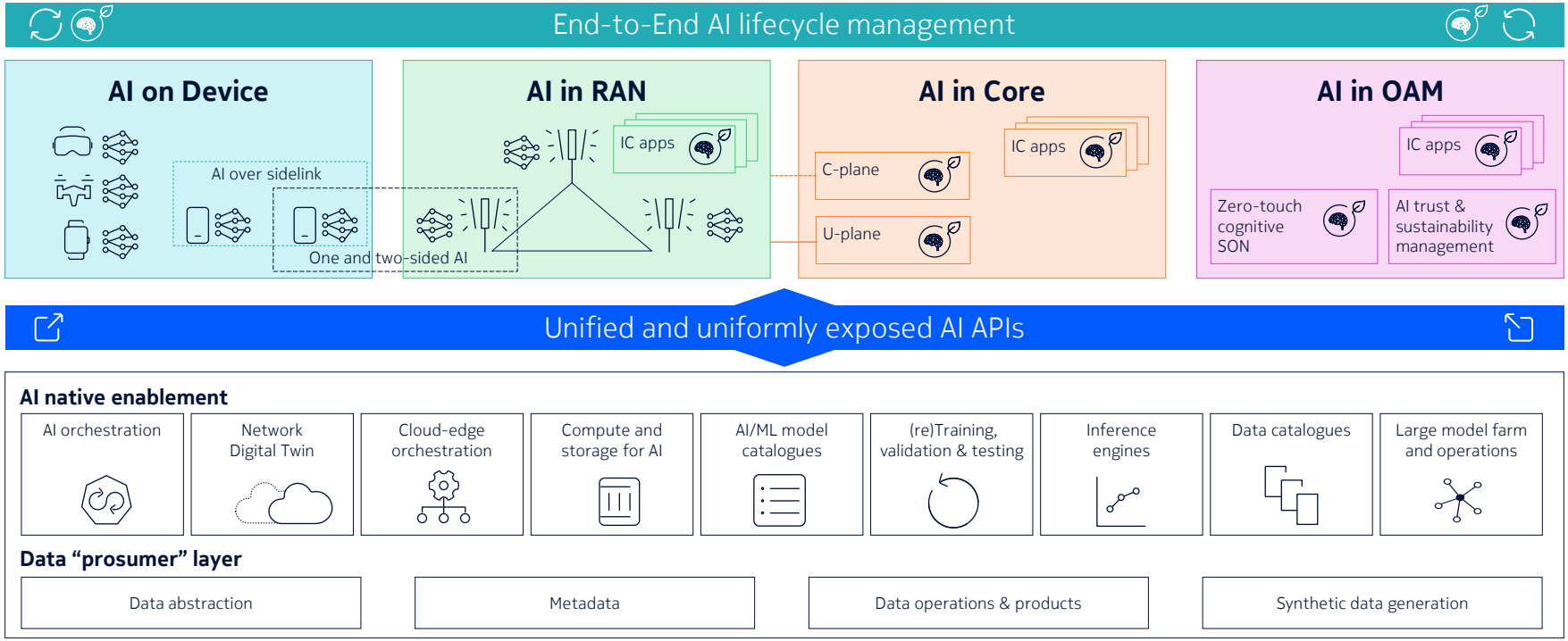
6G presents opportunities to make the network specialized, and the ecosystem programable and monetizable

The **distribution/partitioning** of the data, the cloud resources, the exposure framework and M&O are the choice of the implementation.

 © 2024 Nokia

Framework for AI native

A holistic view



IC: Intelligent Controller

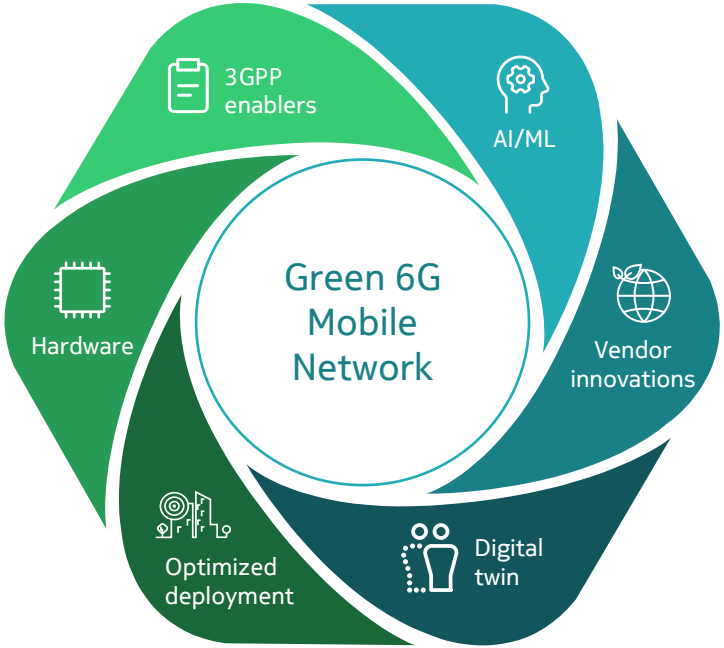
© 2024 Nokia

Framework for energy efficiency

- Apply 5G-Advanced energy saving techniques across all 6G devices from Day One
- Control-and user-plane energy-saving enhancements

- Power-efficient hardware components and product design

- Energy-aware optimizations for coverage, capacity, interference and mobility



- Predicting traffic, radio condition and mobility
- Energy-saving algorithms selection and optimization

- Green-energy site solutions (virtual power plant, liquid cooling, etc)
- Smart workload balancing

- Optimizing energy-saving algorithms and minimizing lab and field-testing efforts

© 2024 Nokia

Ubiquitous 6G service through Non-Terrestrial Networks (NTN)

Natively included for all device types without dependence on external GNSS

NTN support for all UE types
- From 6G LPWA to smartphone and CPE
- Extreme coverage for smartphones by falling back to 6G LPWA

GNSS independent access
- 6G NTN to be independent from external GNSS

TN - NTN Interworking
- Seamless mobility between TN and NTN

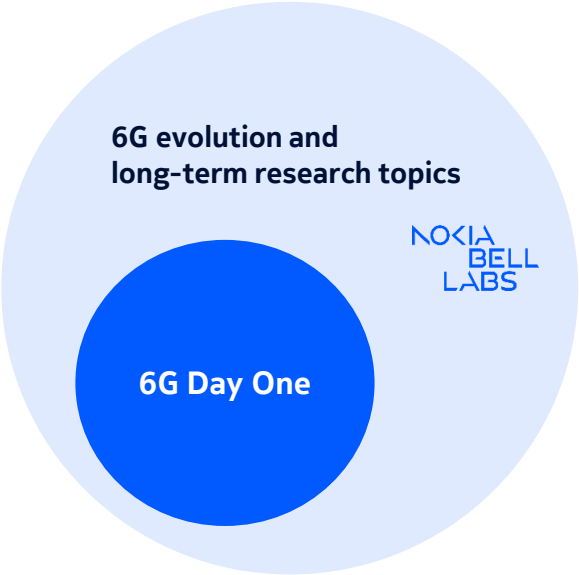
Architecture
- Support for transparent and regenerative architectures.

Multi Orbit support
- GEO – LEO – HAPS networks

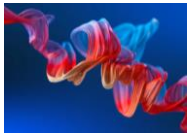
 © 2024 Nokia

6G evolution and beyond

To achieve the full realization of digital-physical fusion



Accelerate digital-physical fusion through Integrated communications and sensing



Harness the full potential of AI to create learning networks that can understand their surroundings and human intent

Transform networking systems and software to a Unified Networking Experience (UNEXT)

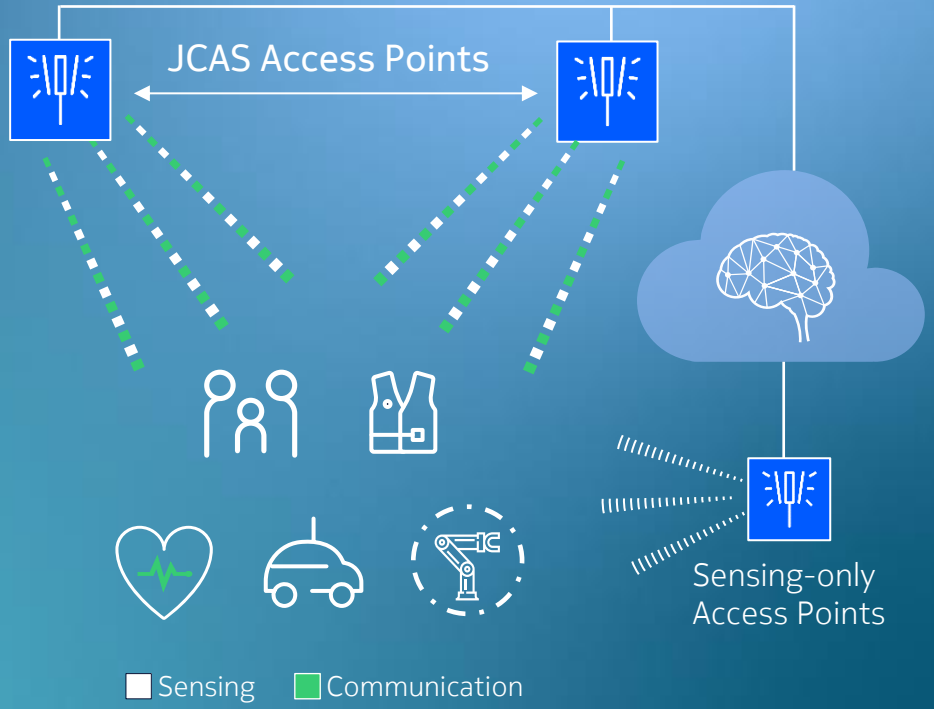
Take energy efficiency to the next level by researching energy-efficient ASIC technology

Open up new spectrum frontiers in the SubTHz bands to augment our access, backhaul and sensing capabilities

© 2024 Nokia

Network as a sensor

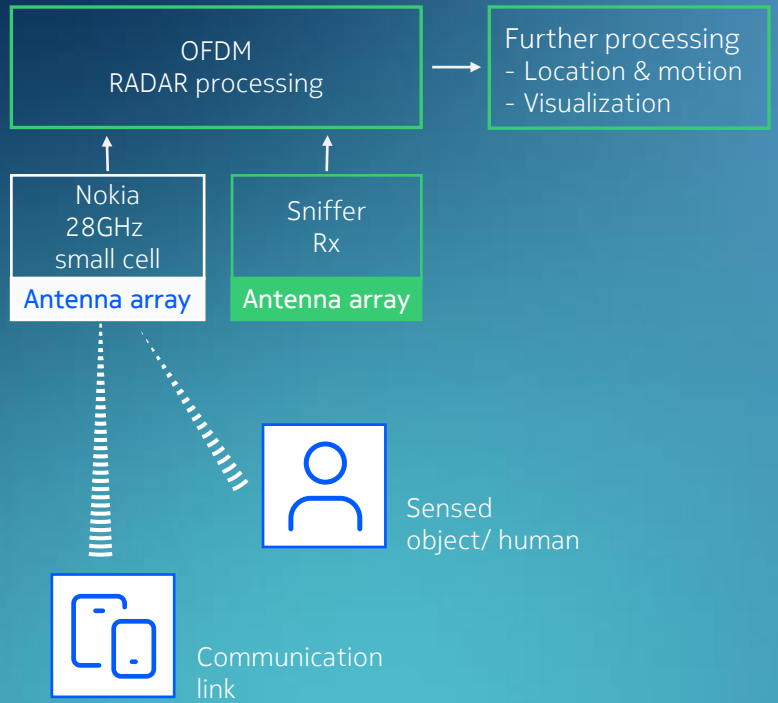
Ubiquitous 6G radio as a new sensing modality

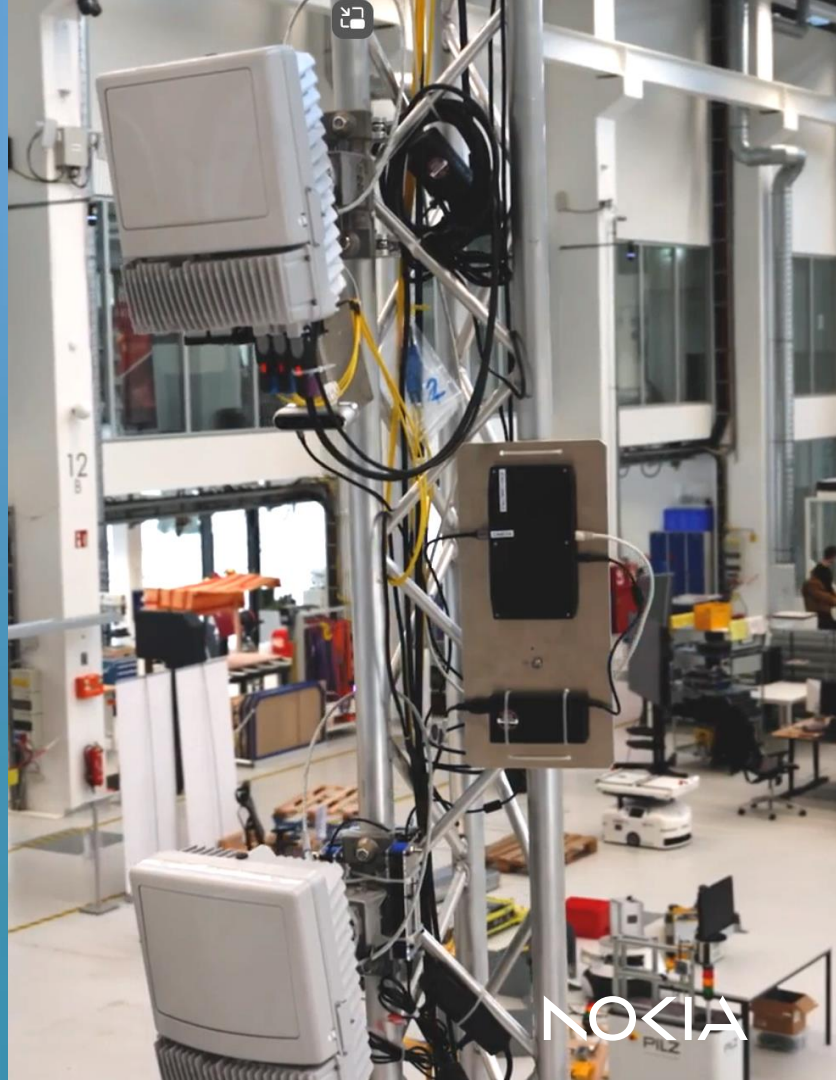


© 2024 Nokia

Network as a sensor

Proof of concept

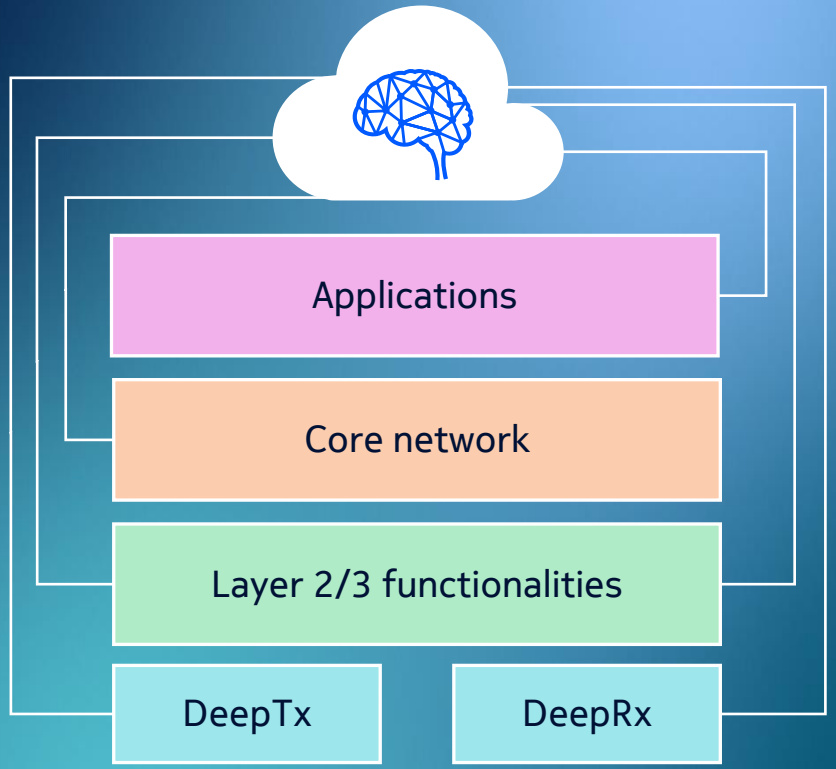




 © 2024 Nokia

The learning network

Build on the foundations laid in 6G Day One to harness the full potential of AI in radio networks



 © 2024 Nokia

The learning network

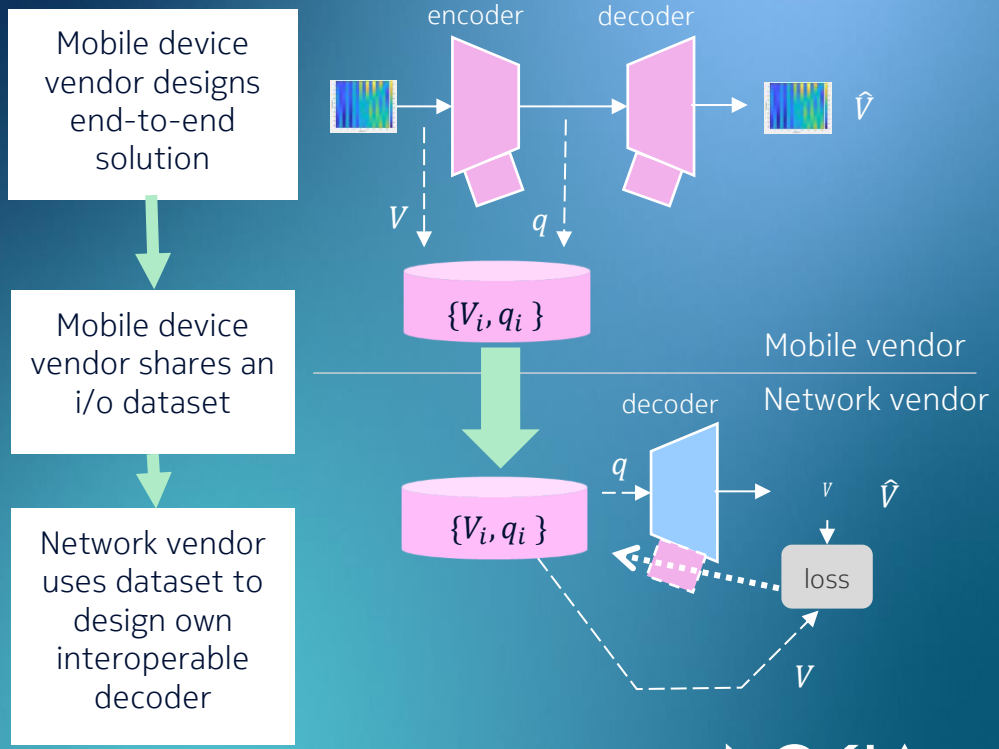
Wireless AI interoperability using sequential training

Interoperability achieved by sharing datasets rather than architectures

- Maintains privacy
- Allows design flexibility

Implementation efficiency

- Can combine datasets to train common decoders compatible with multiple parties



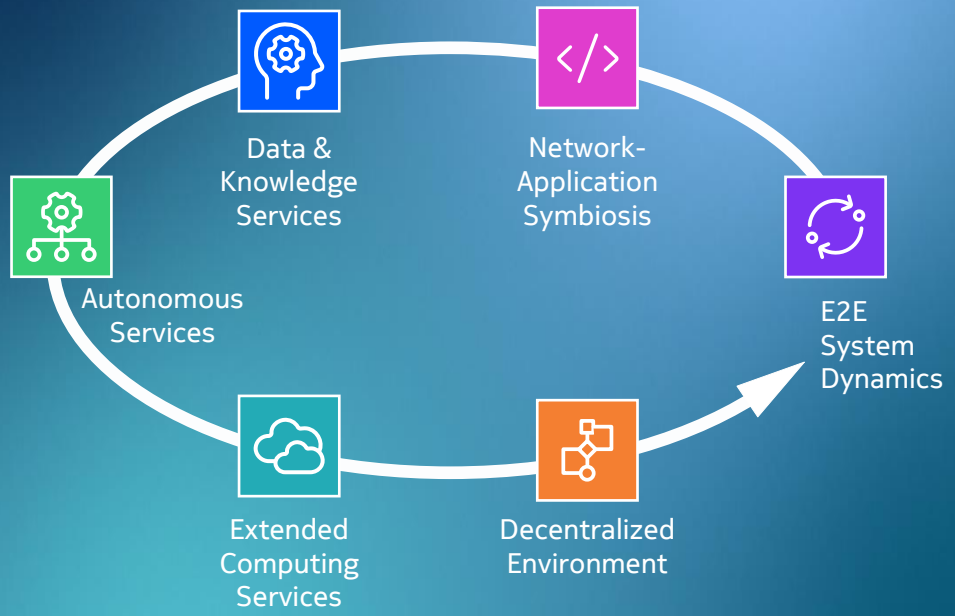
 © 2024 Nokia

UNEXT

Unified Networking Experience Technology

A holistic-system approach to maximize the value of the network

- ✓ Simplicity
- ✓ Security
- ✓ Consumability
- ✓ Sustainability

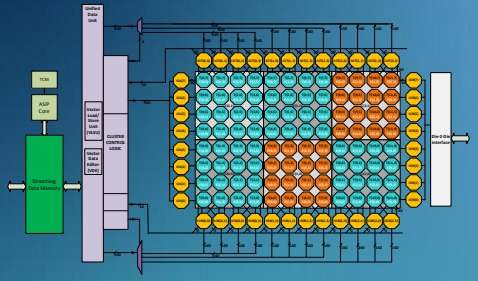


 © 2024 Nokia

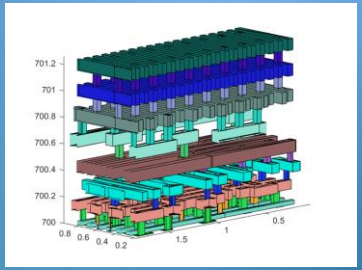
Energy efficient Application-Specific Integrated Circuit (ASIC) technology

As wireless-access compute requirements is outpacing complementary metal oxide semiconductor (CMOS) density evolution, we need new innovations to address it.

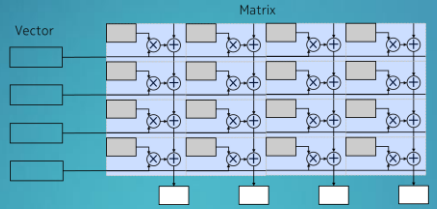
Massive Parallelization for MIMO Wavefront Processing



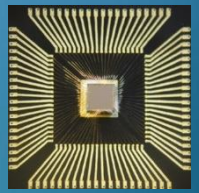
Deep IC Design for Ultra-low Power Analog/Digital Compute



Breaking Memory Wall for AI Compute in Memory



Test-chip Runs for De-risking and to Drive Foundry-related Innovations



 © 2024 Nokia

Shaping the 6G Future

1	Key trends shaping the ecosystem
2	Why 6G, and its design principles
3	6G vision, use cases and technology enablers
4	**Building a tangible foundation for 6G**

Building a tangible foundation for 6G

Following several years of research, the journey towards 6G is becoming clearer

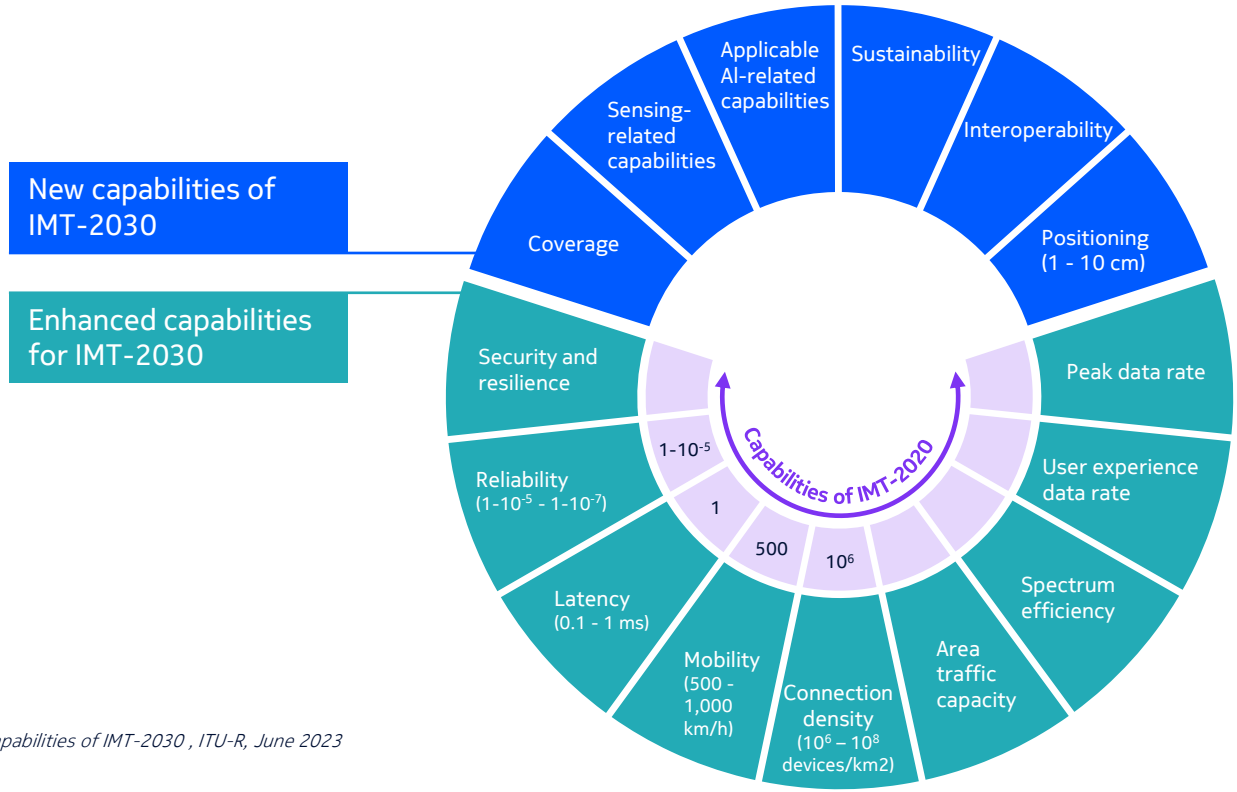
Requirements & evaluation

Spectrum studies

Timeline commitment

© 2024 Nokia

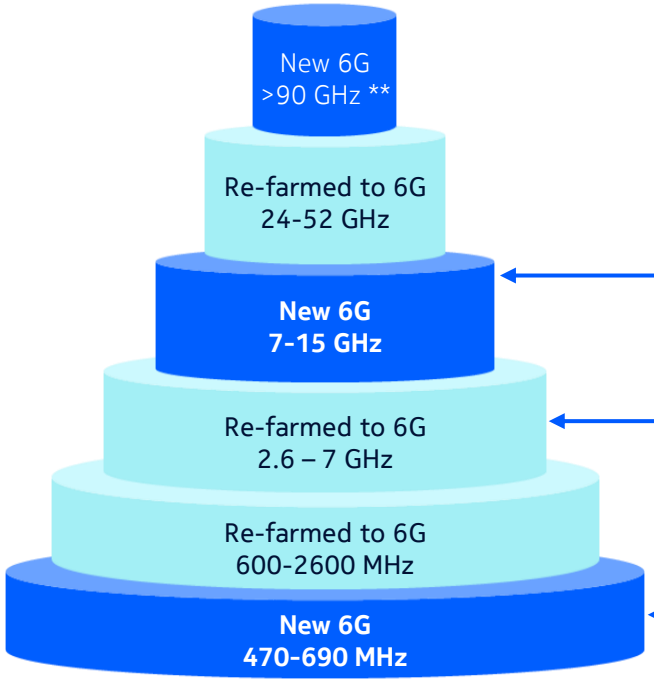
ITU-R has laid out requirements for 6G



Source: Capabilities of IMT-2030 , ITU-R, June 2023

 © 2024 Nokia

WRC-23 lays the groundwork for building 6G systems



Wide area capacity and data rate boost

14.800 GHz	**550 MHz**	15.350 GHz *
7.125 GHz	**1275 MHz**	8.400 GHz *

Expansion of existing mid-bands

6.425 GHz	**700 MHz**	7.125 GHz
4.400 GHz	**400 MHz**	4.800 GHz *

Coverage at low bands **

470 MHz	**220 MHz**	690 MHz

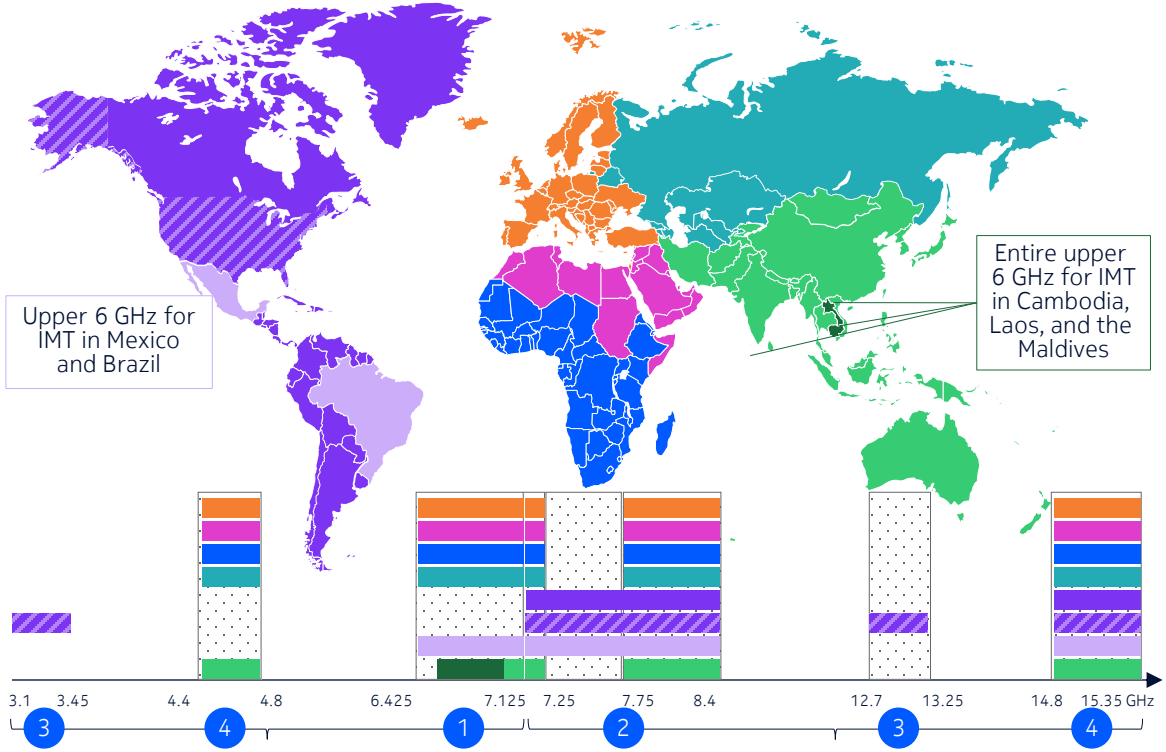
Note: only part of these spectrum blocks will be possible in each geographical regions

* For study towards WRC-27

** Initial proposal to be considered for WRC-31

 © 2024 Nokia

Mid-band spectrum roadmap towards 6G



1. The potential shared use of the Upper 6 GHz band with Wi-Fi could reduce the value of the band for 5G-Adv/6G deployments
2. 7.125-8.4 GHz has great potential to expand mid-band spectrum while enabling macro deployment for increased capacity and coverage
3. In the US, the extension of the C-band towards 3.1 GHz as well as using the 12.7 GHz band will spur additional 6G growth, with the potential to expand the ecosystem beyond US boundaries
4. The 14GHz band is less attractive as it has characteristics closer to mmWave. The 4GHz band is challenging due to existing incumbents (e.g., military use in many countries).

 © 2024 Nokia

Creating the pre-standardized platform that will become the basis for future 6G standardization

European flagship project Hexa-X II

Funded through **Horizon Europe SNS-JU**, with **44 partners** covering the entire value stack, including hardware, system, platform, applications, service providers and academia.

Overall project lead

Driving North American leadership for Next G technologies

A comprehensive model built on North American 6G technology developments, R&D needs, standards goals and market readiness. The Alliance has **over 70 members** and is growing.

Founding member

© 2024 Nokia

3GPP confirms 6G will come with Rel-21

Commercialization will start Q4 2029

1	2	3	4	5
3GPP TSG joint plan December 2023	3GPP 6G workshop March 2025	Rel-21 work item starts March 2027	First 3GPP 6G specifications December 2028	6G commercial launch Q4 2029
Reached agreement on 6G timeline	6G requirements and targets	Key architecture and physical layer decisions	Specs content frozen	6G Day One

Standardisation

Consensus building | 6G requirements and technology studies | 6G standardization | 6G standardization v2

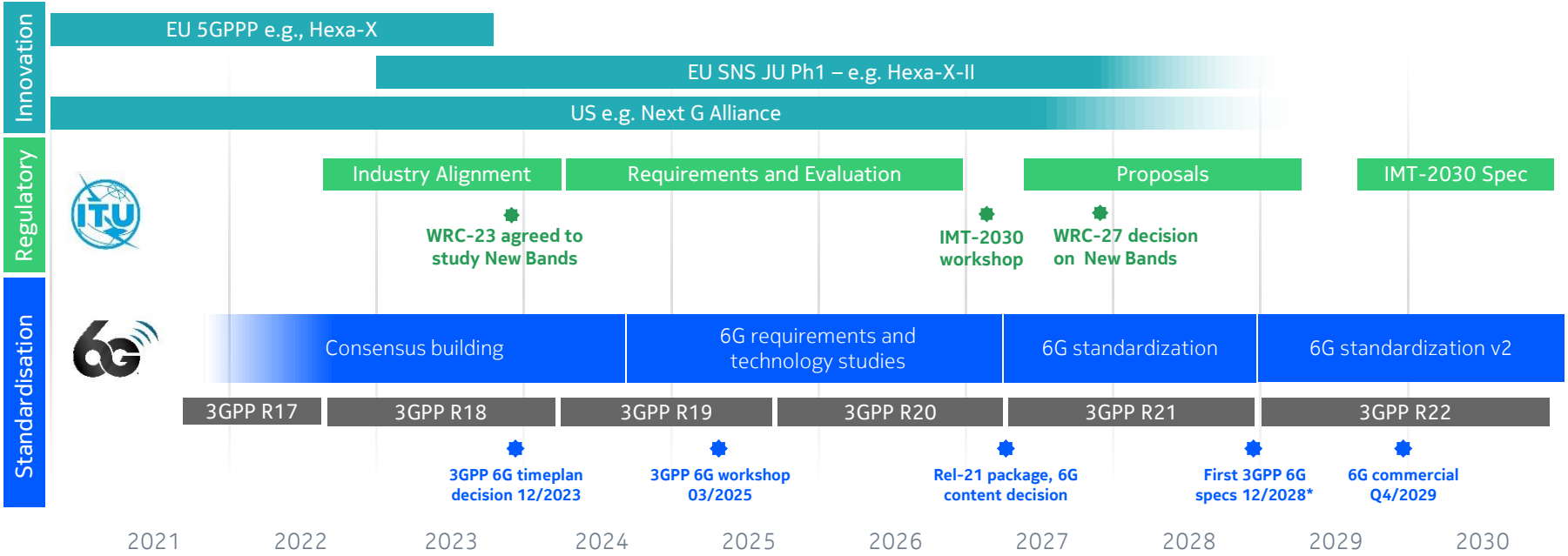
3GPP R17 | 3GPP R18 | 3GPP R19 | 3GPP R20 | 3GPP R21 | 3GPP R22

2021 2022 2023 (1) 2024 2025 (2) 2026 2027 (3) 2028 (4) 2029 (5) 2030

© 2024 Nokia

6G success depends on a global, unified approach

Regulatory, innovation and standardization timelines must be brought into harmony



*3GPP plenary#102 alignment in December 2023

© 2024 Nokia

6G is vital to shaping the future of communications

6G is both evolutionary and revolutionary

Nokia is leading the charge

© 2024 Nokia

NOKIA